

TRABZON, Turkey

WMO: 170380

Lat: 40.995N

Lon: 39.783E

Elev: 34

StdP: 100.92

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.1	1.8	-7.4	2.0	8.2	-5.3	2.4	8.2	12.0	10.5	10.5	11.3	3.5	240	0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.5	28.9	23.9	28.1	23.4	27.2	22.9	25.1	28.0	24.3	27.4	23.7	26.8	3.1	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.0	19.0	27.7	23.1	18.0	27.1	22.4	17.2	26.5	76.8	28.0	73.8	27.6	71.1	26.9	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.0	6.5	-0.9	31.5	1.6	2.1	-2.1	32.9	-3.0	34.2	-3.8	35.3	-5.0	36.8		
(5)			-2.7	25.7	1.4	1.0	-3.8	26.4	-4.6	27.0	-5.4	27.6	-6.4	28.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.6	8.2	8.0	9.3	12.1	16.4	21.1	24.0	24.8	21.6	17.6	13.3	10.1		
	DBStd	6.72	3.68	3.85	4.04	3.63	2.96	2.21	1.65	1.67	2.31	2.92	3.40	3.68		
	HDD10.0	290	79	79	63	15	0	0	0	0	0	1	9	44		
	HDD18.3	1621	313	289	282	192	74	4	0	0	4	49	157	257		
	CDD10.0	2328	25	24	42	77	198	332	436	459	349	235	106	46		
	CDD18.3	617	0	0	2	4	14	86	177	201	103	26	4	1		
	CDH23.3	3159	0	2	21	42	29	230	1024	1412	334	49	14	2		
	CDH26.7	332	0	0	6	11	5	10	94	179	21	6	1	0		
(14) Wind	WSAvg	2.6	2.9	2.9	2.8	2.5	2.3	2.4	2.5	2.6	2.8	2.6	2.7	2.8		
(15) Precipitation	PrecAvg	771	71	56	54	58	52	53	34	43	71	107	90	83		
	PrecMax	911	158	142	107	108	96	148	118	87	150	293	202	174		
	PrecMin	576	13	11	14	25	16	2	12	3	8	4	19	21		
	PrecStd	96	39	34	27	24	20	31	22	25	38	65	47	38		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.0	22.9	25.9	28.1	26.7	27.7	29.2	30.0	28.8	27.2	25.2	22.8		
		MCWB	10.6	11.7	12.5	16.2	18.5	22.5	24.4	25.2	22.7	17.8	13.3	12.1		
	2%	DB	18.4	19.2	21.8	23.2	23.2	26.6	28.2	29.0	27.0	24.4	22.2	19.9		
		MCWB	9.1	9.8	12.0	14.9	18.7	22.3	23.9	24.6	22.4	18.7	13.2	11.1		
	5%	DB	16.1	17.0	18.1	19.3	21.8	25.6	27.8	28.2	26.0	23.0	19.9	17.3		
		MCWB	8.4	9.9	10.9	13.6	18.1	21.7	23.6	23.9	21.8	18.7	13.9	10.7		
	10%	DB	13.2	13.8	15.0	17.0	20.5	24.8	27.0	27.8	25.1	21.9	18.0	15.1		
		MCWB	8.6	8.8	10.3	13.0	17.4	21.1	23.1	23.6	21.2	18.2	13.4	10.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	13.1	14.9	19.4	20.9	24.1	25.9	26.1	24.1	21.2	17.8	14.3		
		MCDB	17.1	19.0	21.7	25.5	24.3	26.7	28.5	28.7	27.1	24.1	21.3	18.8		
	2%	WB	10.6	11.3	12.6	15.4	19.5	23.0	25.0	25.3	23.1	20.2	15.9	12.9		
		MCDB	14.8	17.3	19.2	20.8	22.0	25.7	27.7	28.2	26.2	22.9	18.6	16.3		
	5%	WB	9.9	10.1	11.5	14.2	18.5	22.1	24.1	24.7	22.3	19.3	14.9	11.9		
		MCDB	13.3	14.5	16.7	18.1	21.0	25.0	27.1	27.8	25.4	22.1	17.9	15.4		
	10%	WB	9.1	9.2	10.5	13.2	17.7	21.2	23.4	24.1	21.6	18.4	14.0	11.0		
		MCDB	12.7	13.0	14.5	16.6	20.1	24.1	26.6	27.3	24.7	21.3	17.0	14.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	5.7	6.0	5.8	4.9	4.9	4.7	4.5	5.1	5.1	5.7	5.5		
		MCDBR	8.9	10.4	11.9	11.1	7.3	5.8	5.1	4.8	5.8	6.8	8.5	8.9		
	5% WB	MCWBR	4.4	4.4	4.9	5.7	5.0	4.5	4.1	3.9	4.7	5.1	5.3	4.8		
		MCWBR	7.0	8.8	10.1	9.3	6.1	5.1	4.7	4.3	5.5	6.0	6.6	6.9		
(40) Clear-Sky Solar Irradiance	taub	taub	0.371	0.411	0.472	0.524	0.490	0.460	0.477	0.497	0.439	0.441	0.383	0.361		
		taud	2.240	2.122	1.988	1.904	2.068	2.219	2.208	2.137	2.283	2.184	2.302	2.303		
	Ebn at Noon	Ebn at Noon	755	775	768	757	798	823	804	773	796	735	738	735		
		Edh at Noon	97	126	161	188	163	140	141	147	119	118	91	85		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.84	2.69	3.66	4.61	5.55	6.30	6.17	5.44	4.51	3.00	2.10	1.58		
	RadStd	RadStd	0.19	0.31	0.31	0.48	0.47	0.42	0.41	0.36	0.34	0.28	0.27	0.21		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.63	N/A	-1.81	N/A	N/A	N/A	-65	-143	+175	+94
(47) Regional (4 neighbors)		N/A	N/A	N/A	+0.65	N/A	N/A	N/A	-113	+93	+41

Nomenclature: See separate page